

QUESTION BANK

1. Expand $f(x) = 1 - x^2$ as a Fourier series in the interval $-1 < x < 1$

$$\text{Ans: } a_0 = \frac{4}{3}, a_n = \frac{4}{\pi^2 n^2} (-1)^{n+1}, 1 - x^2 = \frac{2}{3} + \frac{4}{\pi^2} \sum_{n=1}^{\infty} \frac{1}{n^2} (-1)^{n+1} \cos n\pi x$$

2. Find the half range Fourier cosine series for the function $f(x) = x(\pi - x)$ over the

interval $(0, \pi)$. Hence deduce that $\sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{n^2} = \frac{\pi^2}{12}$

$$\text{Ans: } a_0 = \frac{\pi^2}{3}, a_n = \frac{-2}{n^2} [1 + (-1)^n], x(\pi - x) = \frac{\pi^2}{6} - 2 \sum_{n=1}^{\infty} \frac{1}{n^2} [1 + (-1)^n] \cos nx, \text{ put } x = \frac{\pi}{2},$$

we get the required series .

3. Expand $f(x) = \begin{cases} x, & \text{in } 0 < x < \pi/2 \\ \pi - x, & \text{in } \pi/2 < x < \pi \end{cases}$ in half range Fourier cosine series.

$$\text{Ans: } a_0 = \frac{\pi}{2}, a_n = \frac{2}{\pi n^2} \left[2 \cos\left(\frac{n\pi}{2}\right) - 1 - (-1)^n \right]$$

4. Find the Half range sine series for $f(x) = \begin{cases} \frac{1}{4} - x & \text{in } \left(0, \frac{1}{2}\right) \\ x - \frac{3}{4} & \text{in } \left(\frac{1}{2}, 1\right) \end{cases}$

$$\text{Ans: } b_n = \frac{1}{2n\pi} [1 - (-1)^n] - \frac{4}{n^2 \pi^2} \sin \frac{n\pi}{2}$$

5. Find the Half range sine series for $f(x) = (x-1)^2$ in the interval $0 \leq x \leq 1$

$$\text{Ans: } b_n = \frac{2}{n\pi} \left[1 + \frac{2}{n^2 \pi^2} \{(-1)^n - 1\} \right]$$

6. Find the Fourier series for the function $f(x) = x - x^2$ in $-\pi \leq x \leq \pi$.

$$\text{Ans: } a_0 = -\frac{2\pi^2}{3}, a_n = -\frac{4(-1)^n}{n^2}, b_n = -\frac{2(-1)^n}{n}$$

7. Find the Fourier series for the function $f(x) = \left(\frac{\pi - x}{2}\right)^2$ in $0 < x < 2\pi$ and hence deduce that

$$\sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{n^2} = \frac{\pi^2}{12}$$

$$\text{Ans: } a_0 = \frac{\pi^2}{6}, a_n = \frac{1}{n^2}, b_n = 0, \text{ put } x = \pi$$

8. Find the complex Fourier series for the function $f(x) = \cos 2x$ in the interval $-\pi < x < \pi$

$$\text{Ans: } \frac{2}{\pi} \sin(2\pi) \sum_{n=-\infty}^{\infty} \frac{(-1)^n}{4 - n^2} e^{inx}$$

9. Find the complex Fourier series for the function $f(x) = \begin{cases} 0, & \text{for } 0 < x < l \\ 1, & \text{for } l < x < 2l \end{cases}$

$$\text{Ans: } \frac{1}{2} - \frac{1}{\pi} \sum_{n=-\infty}^{\infty} \frac{1}{(2n-1)} \left[e^{\left(\frac{(2n-1)\pi}{l}\right)ix} - e^{-\left(\frac{(2n-1)\pi}{l}\right)ix} \right]$$

10. Find the Fourier series up to first harmonic for $f(x)$ given by the following table:

x°	0	60	120	180	240	300	360
$f(x)$	7.9	7.2	3.6	0.5	0.9	6.8	7.9

$$\text{Ans: } f(x) = \frac{8.9667}{2} + [(4.05)\cos x + (0.8949)\sin x]$$

11. Find the Fourier series for the function $f(x) = \begin{cases} \pi x & 0 \leq x \leq 1 \\ \pi(2-x) & 1 \leq x \leq 2 \end{cases}$

$$\text{Ans: } a_0 = \pi, a_n = \frac{2}{n^2\pi} [(-1)^n - 1], b_n = 0$$

12. Develop $f(x) = e^{-x}$ in Fourier series in the interval $-3 < x < 3$

$$a_0 = \frac{2 \sinh 3}{3}, a_n = \frac{6 \sinh 3 \cos n\pi}{9 + n^2\pi^2}, b_n = \frac{2n\pi \sinh 3 \cos n\pi}{9 + n^2\pi^2},$$

13. Expand $f(x) = \begin{cases} 0 & \text{for } -2 < x < 0 \\ 1 & \text{for } 0 < x < 2 \end{cases}$ in Fourier series.

$$a_0 = 1, a_n = 0, b_n = \frac{1}{n\pi} [1 - (-1)^n]$$

14. Find the Fourier series for the function $f(x) = x - x^2$ in $-1 \leq x \leq 1$.

$$\text{Ans: } a_0 = -\frac{2}{3}, a_n = -\frac{4(-1)^n}{n^2\pi^2}, b_n = \frac{2(-1)^{n+1}}{n\pi}$$

15. Find the Fourier series for the $f(x) = e^x$ in the interval $0 < x < 2\pi$

$$a_0 = \frac{1}{\pi} [e^{2\pi} - 1], a_n = \frac{1}{\pi} \left[\frac{e^{2\pi} - 1}{1 + n^2} \right], b_n = -\frac{n}{\pi} \left[\frac{e^{2\pi} - 1}{1 + n^2} \right],$$

16. Find the Fourier series for the function $f(x) = |\sin x|$ in $-\pi \leq x \leq \pi$.

$$\text{Ans: } a_0 = \frac{4}{\pi}, a_n = -\frac{2}{\pi(n^2 - 1)} [1 + (-1)^n], b_n = 0$$

17. Find the Fourier series for the function $f(x) = x^2$ in $-1 \leq x \leq 1$.

$$\text{Ans: } a_0 = \frac{2}{3}, a_n = \frac{4}{n^2\pi^2} \cos n\pi, b_n = 0$$

18. Expand $f(x) = x \sin x$ as a Fourier series in the interval $0 < x < 2\pi$

$$\text{Ans: } a_0 = -2, a_1 = -\frac{1}{2}, a_n = \frac{2}{n^2 - 1} (n \neq 1), b_n = 0 (n \neq 1), b_1 = \pi$$

19. Obtain a Fourier series to represents the following periodic function

$$f(x) = \begin{cases} 0 & 0 < x < \pi \\ 1 & \pi < x < 2\pi \end{cases}$$

$$\text{Ans: } a_0 = 1, a_n = 0, b_n = -\frac{1}{n\pi} [1 - (-1)^n]$$

20. Find the Fourier series for the $f(x) = e^{-x}$ in the interval $0 < x < 2\pi$

$$\text{Ans: } a_0 = \frac{1}{\pi} [1 - e^{2\pi}], a_n = \frac{1}{\pi} \left[\frac{1 - e^{2\pi}}{1 + n^2} \right], b_n = \frac{n}{\pi} \left[\frac{1 - e^{2\pi}}{1 + n^2} \right],$$

21. Find the Fourier series for the $f(x) = e^{-ax}$ in the interval $-\pi < x < \pi$

$$\text{Ans: } a_0 = \frac{2 \sinh a\pi}{\pi a}, a_n = \frac{2a(-1)^n \sinh a\pi}{\pi(n^2 + a^2)}, b_n = \frac{2n(-1)^n \sinh a\pi}{\pi(n^2 + a^2)}$$

22. Express $f(x) = \cos wx$ in $-\pi < x < \pi$ as a Fourier series, where w is constant.

$$\text{Ans: } a_0 = \frac{2}{\pi w} \sin w\pi, a_n = \frac{(-1)^{n+1} 2w \sin w\pi}{\pi(n^2 - w^2)}, b_n = 0$$

23. Obtain a Fourier series for the function

$$f(x) = \begin{cases} -x & -\pi < x < 0 \\ x & 0 < x < \pi \end{cases}$$

Ans: $a_0 = \pi$, $a_n = \frac{2}{n^2\pi}[-1 + (-1)^n]$, $b_n = 0$

24. Find the Fourier series for the $f(x) = x$ in the interval $-\pi < x < \pi$. Draw its graph.

Ans: $a_0 = 0$, $a_n = 0$, $b_n = -\frac{2}{n} \cos n\pi$

25. Express $f(x) = x$ as a Half-range sine series in $0 < x < 2$

Ans: $a_0 = 0$, $a_n = 0$, $b_n = \frac{4}{n\pi}(-1)^n$, $f(x) = \frac{4}{\pi} \sum_{n=1}^{\infty} (-1)^{n+1} \sin \frac{n\pi}{2} x$

26. Express $f(x) = x$ as a Half-range cosine series in $0 < x < 2$

Ans: $a_0 = 2$, $a_n = \frac{4}{n^2\pi^2} [(-1)^n - 1]$, $f(x) = 1 - \frac{8}{\pi^2} \sum_{n=1}^{\infty} \frac{1}{(2n-1)^2} \cos[(2n-1)\frac{\pi x}{2}]$

27. Express $f(x) = x^2$ as a Half-range cosine series in $0 < x < \pi$

Ans: $a_0 = \frac{2}{3}\pi^2$, $a_n = \frac{4}{n^2} [(-1)^n]$

28. Find the Fourier series for the function $f(x) = x^2$ in $-l \leq x \leq l$.

Ans: $a_0 = \frac{2l^2}{3}$, $a_n = \frac{4l^2(-1)^n}{n^2\pi^2}$, $b_n = 0$

29. Find the value of $\sum_{n=1}^{\infty} \frac{1}{n^2}$ using Fourier series. [Assume $f(x) = x^2$ in the interval $(-\pi, \pi)$]

Ans: $\frac{\pi^2}{6}$

30. Find the value of $\sum_{n=1}^{\infty} \frac{1}{4n^2 - 1}$ using the Fourier series. Given $f(x) = \sqrt{1 - \cos x}$ in the interval $0 < x < 2\pi$

Ans: $\sum_{n=1}^{\infty} \frac{1}{4n^2 - 1} = \frac{1}{2}$; $a_0 = \frac{4\sqrt{2}}{\pi}$, $a_n = -\frac{4\sqrt{2}}{\pi(4n^2 - 1)}$, $b_n = 0$

31. Find the Half range Fourier cosine series for the function $f(x) = x - x^2$ in $0 < x < 1$.

Ans: $a_0 = \frac{1}{3}$, $a_n = -\frac{2}{\pi^2 n^2} [(-1)^n + 1]$

32. Obtain a cosine series for $f(x) = \begin{cases} 1 & \text{for } 0 < x < 1 \\ x & \text{for } 1 < x < 2 \end{cases}$

$$\text{Ans: } f(x) = \frac{5}{4} - \frac{4}{\pi^2} \left[\cos\left(\frac{\pi x}{2}\right) - \frac{1}{2} \cos\left(\frac{2\pi x}{2}\right) + \frac{1}{9} \cos\left(\frac{3\pi x}{2}\right) + \dots \right]$$

33. Find the Fourier series for the function $f(x) = 2x - x^2$ in $0 < x < 2$.

$$\text{Ans: } a_0 = \frac{4}{3}, a_n = -\frac{4}{n^2 \pi^2}, b_n = 0$$

34. Find the Fourier series up to first harmonic for $f(x)$ given by the following table:

x	0	1	2	3	4	5
$f(x)$	9	18	24	28	26	20

$$\text{Ans: } f(x) = \frac{41.66}{2} + [(-8.33)\cos x + (-1.15)\sin x]$$

35. Find the Fourier series up to first harmonic for $f(x)$ given by the following table:

x°	0	60	120	180	240	300	360
$f(x)$	0.8	0.6	0.4	0.7	0.9	1.1	0.8

$$\text{Ans: } f(x) = 0.75 + [(0.10)\cos x + (-0.29)\sin x]$$

36. Find the Fourier series up to first harmonic for $f(x)$ given by the following table:

x°	0	60	120	180	240	300	360
$f(x)$	7.9	7.2	3.6	0.5	0.9	6.8	7.9

$$\text{Ans: } f(x) = \frac{8.9667}{2} + [(4.05)\cos x + (0.8949)\sin x]$$

37. The turning moment T units of a crank shaft of a steam engine are given for a series of values of the crank angle θ in degrees.

θ	0	30	60	90	120	150	180
T	0	5224	8097	7850	5499	2626	0

Find the first three terms of sine series represent T

$$\text{Ans: } T = b_1 \sin \theta + b_2 \sin 2\theta + b_3 \sin 3\theta = 7.850 \sin \theta + 1.500 \sin 2\theta + 0$$

38. Find the Fourier series for the function $f(x) = x^2 - 2$ in $-2 < x < 2$.

$$\text{Ans: } a_0 = -\frac{1}{3}, a_n = \frac{16}{\pi^2 n^2} (-1)^n, b_n = 0$$

39. In the range $(-2, 2)$, $f(x)$ is defined by the relation

$$f(x) = \begin{cases} 0, & -2 < x < 0 \\ a, & 0 < x < 2 \end{cases}$$

Expand $f(x)$ in Fourier series.

Ans: $a_0 = a, a_n = 0, b_n = \frac{a}{n\pi} [1 - (-1)^n], f(x) = a \left[\frac{1}{2} + \frac{1}{\pi} \sum_{n=1}^{\infty} \frac{1}{n} \{1 - (-1)^n\} \sin\left(\frac{n\pi x}{2}\right) \right]$

40. Find the Fourier series for the function $f(x) = 2x - x^2$ in $0 < x < 3$.

Ans: $a_0 = 0, a_n = -\frac{9}{n^2 \pi^2}, b_n = \frac{3}{\pi n}$

Ans: $a_0 = -\frac{1}{2}, a_n = \frac{8}{n^2 \pi^2} \left[\cos\left(\frac{n\pi}{2}\right) - 1 \right], b_n = 0$

41. Find the half range cosine series for $f(x) = \begin{cases} c & 0 < x < a \\ 0 & a < x < l \end{cases}$

Ans: $a_0 = \frac{ac}{l}, a_n = \frac{2c}{n\pi} \sin\left(\frac{n\pi a}{l}\right)$

42. Find the value of $1 - \frac{1}{3} + \frac{1}{5} - \frac{1}{7} + \dots$ using Fourier series. Given

$$f(x) = \begin{cases} -1+x, & -\pi < x < 0 \\ 1+x, & 0 < x < \pi \end{cases}$$

Ans: $\frac{\pi}{4}$

43. Find the Fourier series for the function $f(x) = \begin{cases} -1, & -3 < x < 0 \\ 1, & 0 < x < 3 \end{cases}$

Ans: $\frac{\pi}{4} \sum_{n=1}^{\infty} \frac{1}{(2n-1)} \sin\left\{(2n-1)\frac{\pi x}{3}\right\}$

44. Find the Fourier coefficients and Fourier series of the square-wave function f defined by

$$f(x) = \begin{cases} 0 & \text{if } -\pi \leq x < 0, \\ 1 & \text{if } 0 \leq x < \pi \end{cases} \quad \text{and } f(x + 2\pi) = f(x),$$

$f(x)$,

Ans:

$$\frac{1}{2} + \sum_{k=1}^{\infty} \frac{2}{(2k-1)\pi} \sin(2k-1)x$$

45. Find the Fourier series of $f(x) = \frac{\pi-x}{2}$ in $0 < x < 2\pi$ Hence deduce that $1 - \frac{1}{3} + \frac{1}{5} - \frac{1}{7} + \dots = \frac{\pi}{4}$.

Ans: $f(x) = \frac{\pi-x}{2} = \sum_{n=1}^{\infty} \frac{1}{n} \sin nx$

46. Find the complex form of Fourier series for the function $f(t) = \sin t$ in $(0, \pi)$

Ans: $\frac{2}{\pi} \left[1 - \frac{e^{2it} + e^{-2it}}{1.3} - \frac{e^{4it} + e^{-4it}}{3.5} - \frac{e^{6it} + e^{-6it}}{15.7 - \dots} \right]$

47. Find the Fourier series of $f(x) = x(2\pi - x)$ in $0 < x < 2\pi$

Ans:

$$f(x) = \frac{4\pi^2}{6} + \sum_{n=1}^{\infty} \frac{-4}{n^2} \cos nx$$

48. Find the Fourier half range a) Cosine series b) Sine series of

$$f(x) = \begin{cases} x, & 0 < x < 1 \\ 2-x, & 1 < x < 2 \end{cases}$$

Ans: $\frac{8}{n^2\pi^2} \cos\left(\frac{n\pi}{2}\right) - \frac{4}{n^2\pi^2} [1 + (-1)^n], \sum_{n=1}^{\infty} \frac{8}{n^2\pi^2} \sin\left(\frac{n\pi}{2}\right) \sin \frac{n\pi x}{2}$

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